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INTRODUCTION

Online food ordering is the process of ordering food from a website. The product can either be food that has been specially prepared for direct consumption (such as vegetables straight from a farm or garden, frozen meats, etc.) or food that has not been (such as direct from a certified home-kitchen, restaurant). The effort to create an online food ordering system aims to replace the manual method of taking orders with a digital one. The ability to rapidly and correctly create order summary reports whenever necessary is a key factor in the development of this project.

The potential of an online food ordering system is enormous. Any restaurant or fast food chain can use this PHP project to keep track of customer orders. This project is simple, quick, and precise. There is less disk space needed. MYSQL Server is used as the backbone by the online food ordering system, eliminating the risk of data loss and ensuring data security. Customers have the option of either having the food delivered or picked up. A customer starts by selecting the restaurant of their choice, then scans the menu, picks an item, and then decides whether they want it delivered or picked up. Then, when picking up the food, you can pay with cash at the restaurant or with a credit card or debit card using the app or website. The customer is informed by the website and app about the food's quality, how long it takes to prepare, and when it will be ready for pick-up or delivery.

Objectives

The management of the information regarding item category, food, delivery address, order, and shopping cart is the system's primary goal. It oversees the management of all customer, shopping cart, and item category information. Since the project was entirely developed on the administrative end, only the administrator is assured access. The goal is to develop an application program to simplify managing the food consumer item category. It keeps note of every delivery address requested.

Functionalities

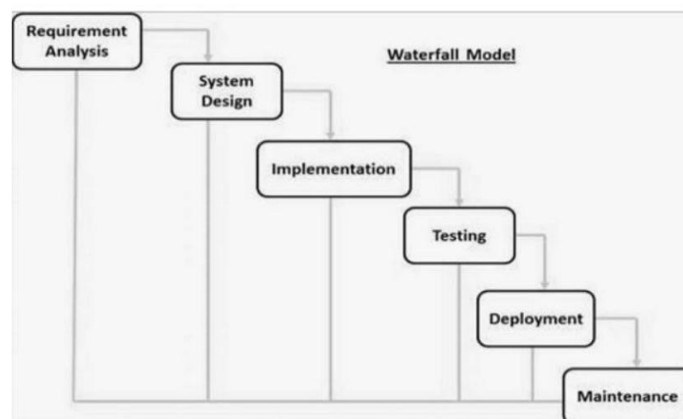
- Provides search options based on a variety of criteria. like Food Item, Customer, Order, and Order Confirmation.
- Online food ordering systems also manage payment information for order details, order confirmation details, and food items online.
- It keeps track of all the data regarding Categories, Payments, Orders, etc.
- Manage the category's details.
- Displays the food item's information and description for the customer
Easy to manage the Food Item, Category more effectively.
- It focuses on keeping track of order's data and transactions.
- Manage the food item's information.
- Improvements in editing, adding, and updating records lead to proper resource management of food item data.
- Manage the order's information by combining all Confirm Order data.

Methodology

Tools and Technique

1. Python
2. Nginx
3. PostgreSQL
4. ReactJs
5. Bootstrap
6. Amazon Cloud
7. Git
8. Java Script
9. HTML

Methodology Development Model

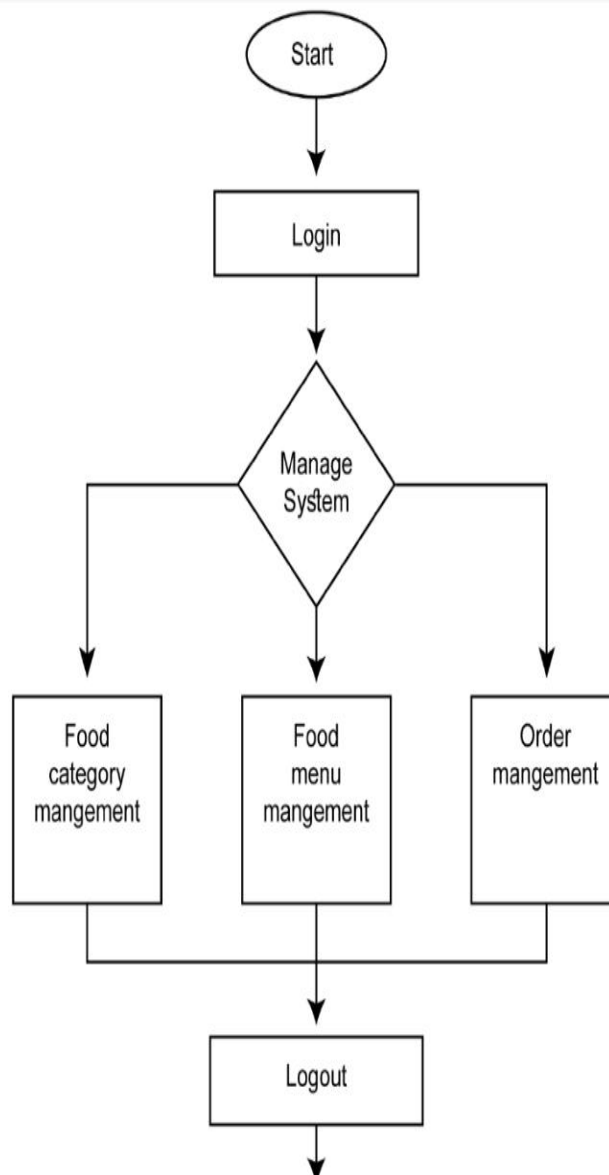


- Requirement Gathering and analysis – During this stage, all potential system needs are gathered and outlined in a requirement specification document.
- System Design – The system design is created in this phase after studying the requirement specifications from the first phase. This system design aids in determining the overall system architecture as well as the hardware and system requirements.
- Implementation – The system is initially built in discrete programs known as units, which are then combined in the following phase, using inputs from the system design. Unit testing is the process of developing and evaluating each unit for functionality.
- Integration and Testing – Following the testing of each unit created during the implementation phase, the entire system is merged. The entire system is tested for errors and failures after integration.

- Deployment of system – Once the product has undergone functional and non-functional testing, it is either published to the market or deployed in the customer's environment.
- Maintenance – Various problems can arise in a client environment. Patches are published to address certain problems. Additionally, improved versions of the product are issued. To bring about these changes in the surroundings of the consumer, maintenance is performed.

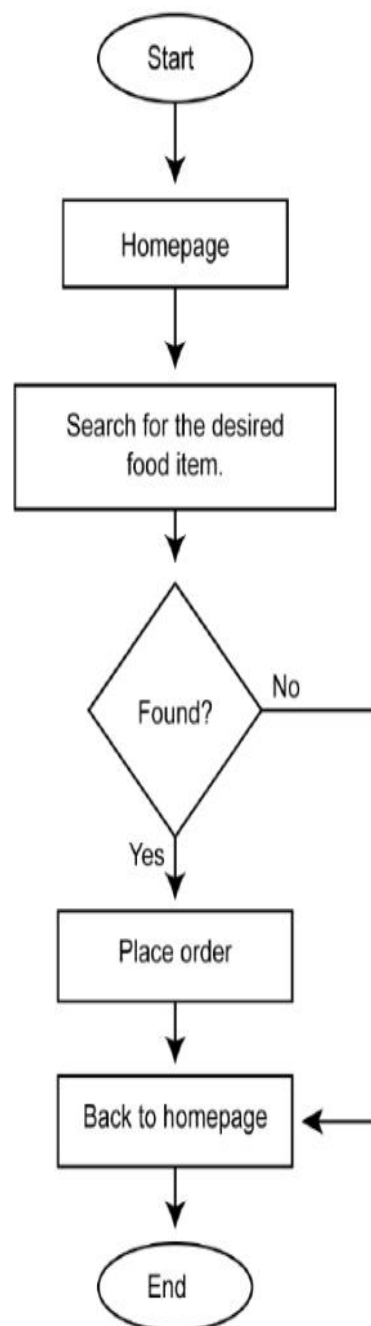
Admin workflow Process

User goes to home page of the domain. If he/she has an account then he/she can login in restaurant management system otherwise he/she need to register an account after successful registration, they can login in home page.

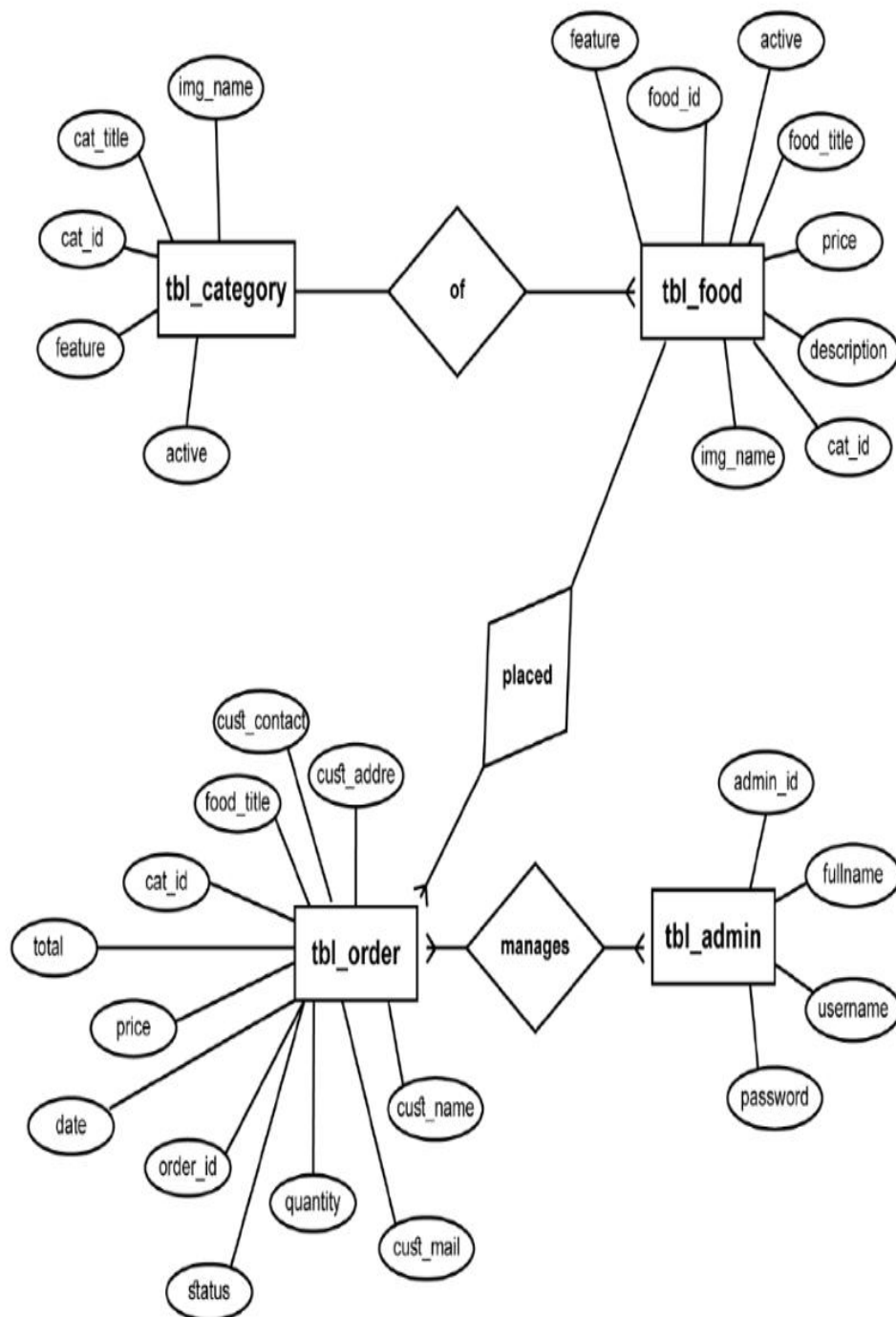


Customer Workflow Process

Initially to visit the food categories or food menu, users don't need to login/register an account. After checking out the categories and menu items, if the user finds his/her desired menu and if they want to order that particular item they can go to order page. During placing any order the customer needs to provide his/her required information mentioned the order section.



E-R Diagram



ANALYSIS RESULT & DISCUSSION

System Implementation Plan

A software design pattern called Model View Controller, or MVC as it is more formally known, is used to build online applications. There are three components to the Model View Controller pattern:

- Model - The lowest level of the pattern, is in charge of maintaining the data.
- View - This is in charge of showing the user all or part of the data.
- Controller - The computer program that controls how the Model and View interact. MVC is well-liked because it provides for duty separation by separating the application logic and user interface layers. The Controller accepts all requests from the application and collaborates with the Model to prepare any necessary data for the View. The View then constructs a final presentable response using the data produced by the Controller. The following is a graphic representation of the MVC abstraction.
Model of MVC (Model View Controller Flow)

Project Planning

Here is an illustration of a software project plan:

- 1) How will the project be carried out within the company? What are the time, financial, and human resource limitations? What does having a market strategy entail?
- 2) Customer meetings: Weekly or as needed customer meetings that include a progress report presentation. Additionally taking into account customer input, adjustments and changes are made as necessary. The client is also shown project deliverable and milestones.

The steps listed below can be used to create successful software projects: Select a project. The aims and objectives of project are as follows:

- Understanding specifications and requirements.
- Using analysis, design, and implementation methods.
- Using testing procedures.
- Documenting.
- Budget allocation or exceeding limits under control.
- Understanding project milestones and deliverable.
- Project estimates.
- Cost and Time.

Facing Problem During Development the Project

During the construction of the web application "Online Food Order," the developer ran into a few issues. Here are a few issues in brief:

- **Requirement Gathering Phase** : It is a crucial step. The project will fail if the requirements are poor. At that time, developer became disappointed when Developer was collecting information and data then what information and data will be helpful or appropriate for this project.
- **During Design Phase** : At this moment, the developer struggled to decide which flowchart would be best for this project when creating it.
- **Development Phase** : It is a very major component of the undertaking. Frequently, the developer misplaced the semicolon (;) at the conclusion of the statement.
- **Testing Phase** : It is an essential component of the project. This section will aid with project testing overall. During testing, developer has faced some bugs of the project.

Limitations of the System

The system has certain other restrictions as well. There are only a few basic functions in the system's shopping cart, and it cannot be extensively customized. Additionally, practically all of the functionality of the application, including validation, is handled by server-side programming. It increases the server's workload, especially when a large number of users access the program. This issue can be resolved by using client-side languages, such as JavaScript or HTML 5, to check data. Additionally, the order model has been created.

Application

- Restaurants, takeaways, and businesses that sell food to go profit from internet meal ordering software designed specifically for them. Customers like the ease of online meal ordering, which is why it is expanding quickly. Expand your sales channels by downloading our online food ordering application.
- Through this food ordering website, customers may place orders from their computers, tablets, and cellphones. They can look through your menu options, choose what they want, and submit an order online. Internet-based payment will also be accepted. Meals can be picked up in person or delivered to customers.
- There are many benefits to using an online food ordering app or a restaurant ordering app, including reduced labor expenses, fewer walk-away customers, and shorter wait times. This restaurant's online ordering system is intended for independent and multi-location chains that offer food to go, including eateries, fast food outlets, take-out, and other catering services
- Putting your company online will enable you to generate a lot more revenue, which will enhance your marketability. Your online menu will give current clients a terrific new option to place orders, and new customers will easily find you thanks to well-known search engines. To complement the style and feel of your present website, the system is tailored. In the digital age, we help business owners grow their enterprises.

Conclusion

Restaurant Management System is a web-based technology that aids the restaurant industry in carrying out tasks effectively and efficiently. It aids in managing cash flow for managers. Managers can view analytics data to assess company growth. The manager can control orders and employee schedules by using this system. The full complement is a restaurant management system. It provides access to the Online Order platform, third-party connectors software, and comprehensive CRM solution, which together cover a sizable portion of your restaurant's requirements. They are not the outdated hardware and software sets for restaurants that were previously offered. They are the hottest things around, smooth, manageable, inexpensive, and quick. In the "Online Food Ordering Project," we made every effort to meet all the demands of the restaurant. Because it is straightforward and adaptable, the project is successful. The biggest benefit of my project is that it draws plenty of users because of its simplicity. A novice user may operate it with ease. Any type of restaurant can utilize our software. By automating meal ordering, billing, and inventory control, the restaurant management system assists the restaurant manager in managing the restaurant more successfully and efficiently. The system handles the transaction and stores the data produced. These data will be used to create reports that assist the restaurant manager in making wise business decisions. For example, the manager can decide whether more waiters, delivery men, delivery carts, and cooks are needed based on how many clients will be present during a specific time period. When this project is finished, all security concerns will be resolved.

Additionally, a quick and secure authentication process will be used for record maintenance. Because it automatically pulls information about a consumer from the database on subsequent visits, data entry is quick and easy. As a result, our program will undoubtedly succeed in replacing the antiquated manual way of storing secure information. The work plan also specifies the specific front end and back end characteristics of the technology being used in the project. Future project goals and its scope have been elaborated.

Future Work

Each project should pay close attention to future development because it contains the system's most recent features. It lessens software issues and defects. It develops a close relationship with customers based on their comments or preferences. Developer will incorporate certain dynamic elements that are briefly described below into my restaurant management system. Reporting module with real time mechanism.

- Modern architecture with smooth transitions.
- System for email and mobile confirmation
- Selling Point